ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY 4025/1

PAPER 1 Multiple Choice

NOVEMBER 2022 SESSION 1 hour

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.
Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 15 printed pages and 1 blank page.

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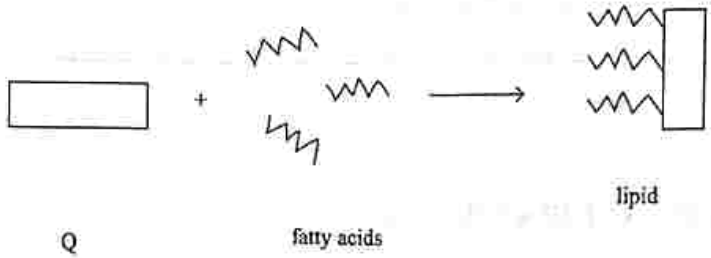
1 Which branch of Biology involves the study of hereditary traits?

A anatomy
B physiology
C genetics
D zoology

2 Which statement is **not** a laboratory safety rule?

A Use equipment for its intended use.
B Tie back long hair and loose clothing.
C Wear safety googles when pouring liquids.
D Place the measuring cylinder on a flat surface.

3 The diagram shows the formation of a lipid.

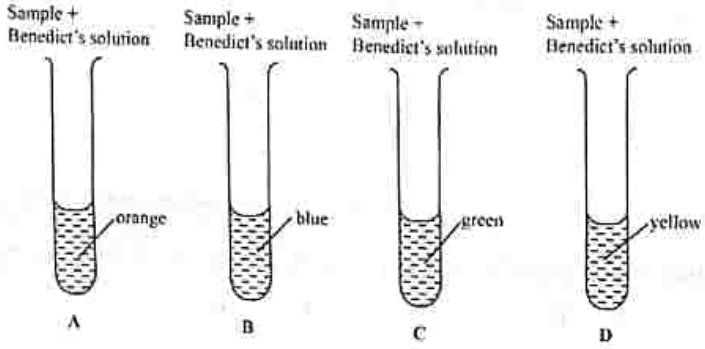


What is Q?

A glucagon
B glucose
C glycogen
D glycerol

4 The diagrams show an experiment to determine nutrients in a food sample.

Which test tube, **A**, **B**, **C** or **D**, had the highest amount of reducing sugar?

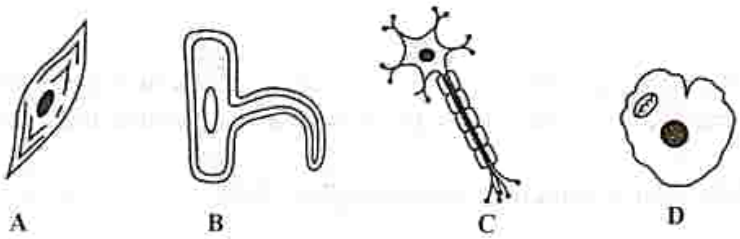


5 Which one is a result of water being polar?

A reagent in photosynthesis
B product of respiration
C high thermal capacity
D most dense at 4°C

6 The diagram shows four specialised cells.

Which cell, **A**, **B**, **C** or **D**, is specialised for contraction?

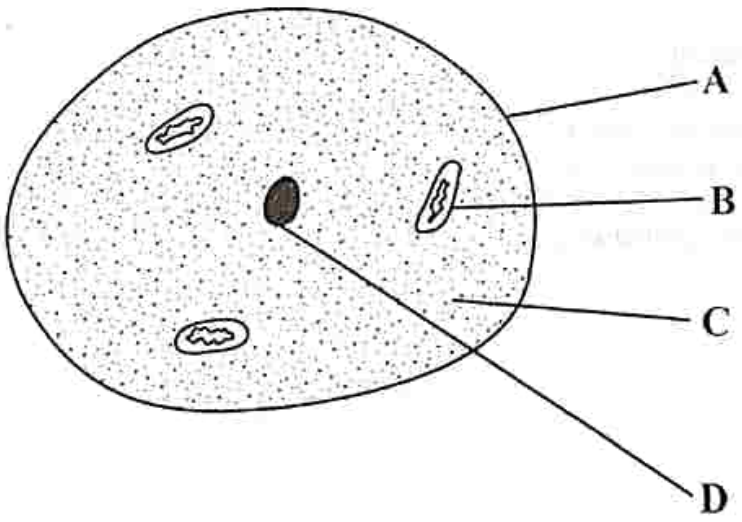


7 Which one is a characteristic of a sperm?

A lobed nucleus
B numerous mitochondria
C granulated cytoplasm
D large size

8 The diagram shows an animal cell.

In which part, **A**, **B**, **C** or **D**, does anaerobic respiration take place?



9 What stops salivary amylase activity in the stomach?

A low enzyme concentration
B low substrate concentration
C low temperature
D low pH

10 Which enzyme should be included in a detergent to remove a cream stain?

A catalase
B lipase
C amylase
D protease

11 A student carried out an experiment using pieces liver, water and hydrogen peroxide, to investigate enzyme activity. Two of the pieces of liver were fresh and two were boiled.

In which test tube did the combination produce bubbles?

A fresh liver and hydrogen peroxide
B boiled liver and hydrogen peroxide
C fresh liver and water
D boiled liver and water

12 Which one is a method of artificial vegetative reproduction in plants?

A suckers
B cuttings
C runners
D bulbs

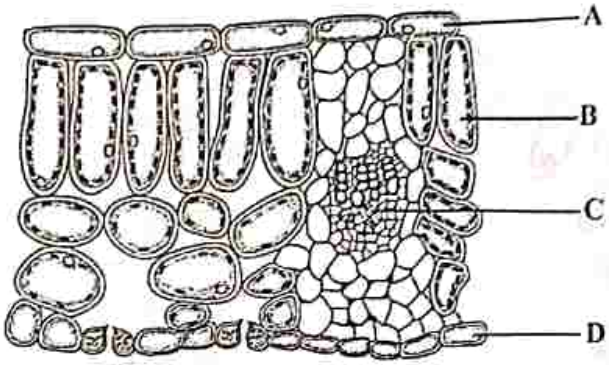
13 A reddish powdery substance was observed on the surfaces of tomato leaves and stems.

This is a sign of

A tissue eating pest.
B sap sucking pest.
C bacterial wilt disease.
D fungal rust disease.

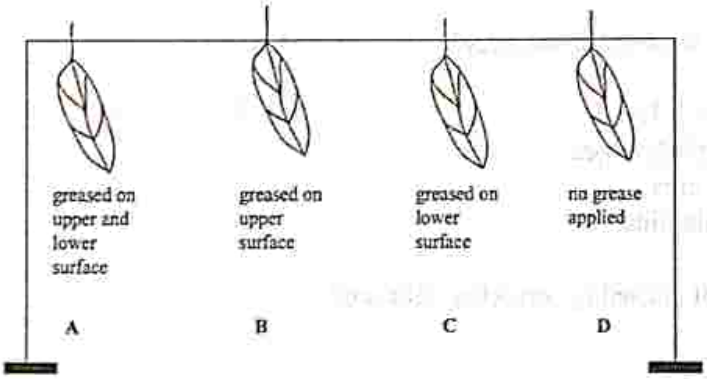
14 The diagram shows the cross section through a dicotyledonous leaf.

Which part, **A**, **B**, **C** or **D**, contains the vascular bundles?



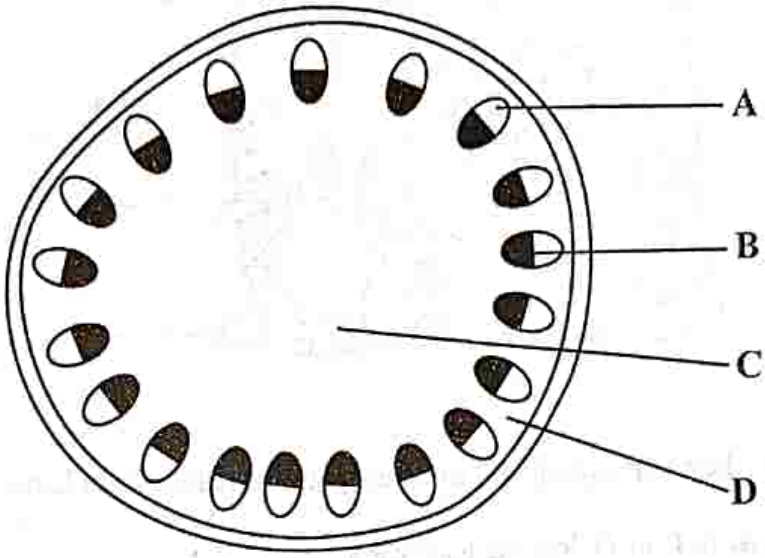
15 The diagram shows an experiment to investigate water loss from leaves.

Which leaf, **A**, **B**, **C** or **D**, lost the least water?



16 The diagram shows a section of a dicotyledonous stem.

Which part, **A**, **B**, **C** or **D**, is responsible for translocation?



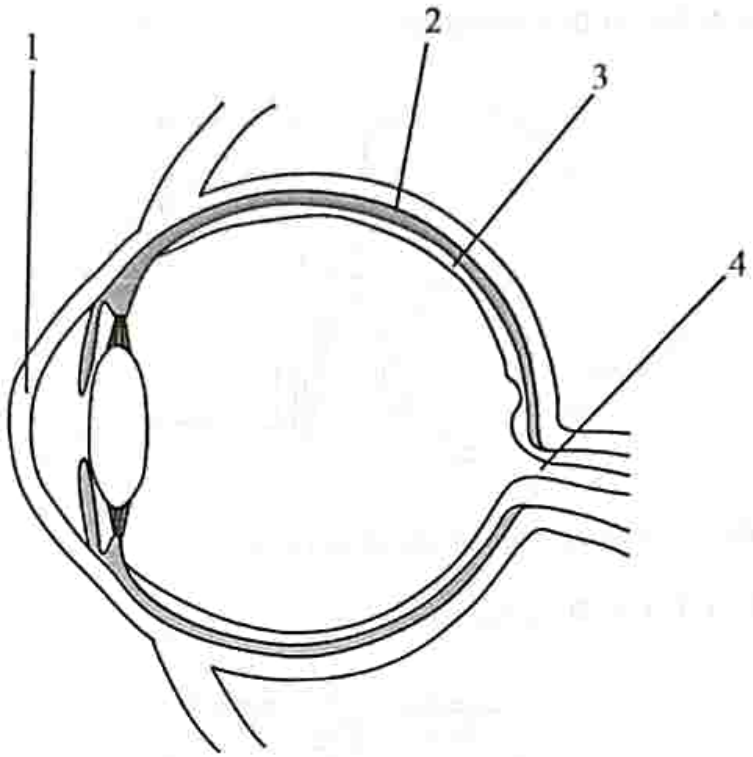
17 Which one is **not** a function of the liver?

A production of bile
B breakdown of glycogen
C excretion of urea
D storage of vitamins

18 Which disease is **not** caused by smoking tobacco?

A bronchitis
B pneumonia
C lung cancer
D emphysema

19 The diagram shows a section of the human eye.

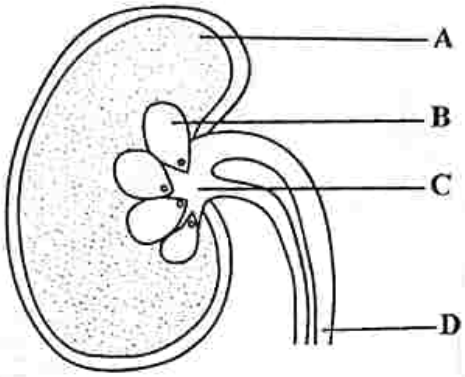


Which numbered part is light sensitive?

A 1
B 2
C 3
D 4

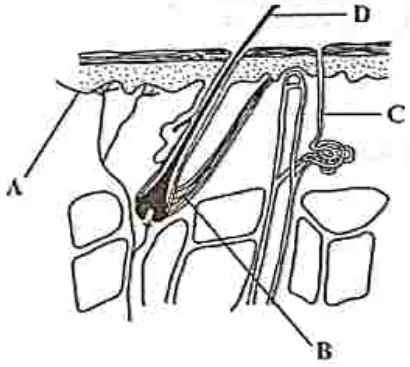
20 The diagram shows the human kidney.

Which part, **A**, **B**, **C** or **D**, is the cortex?



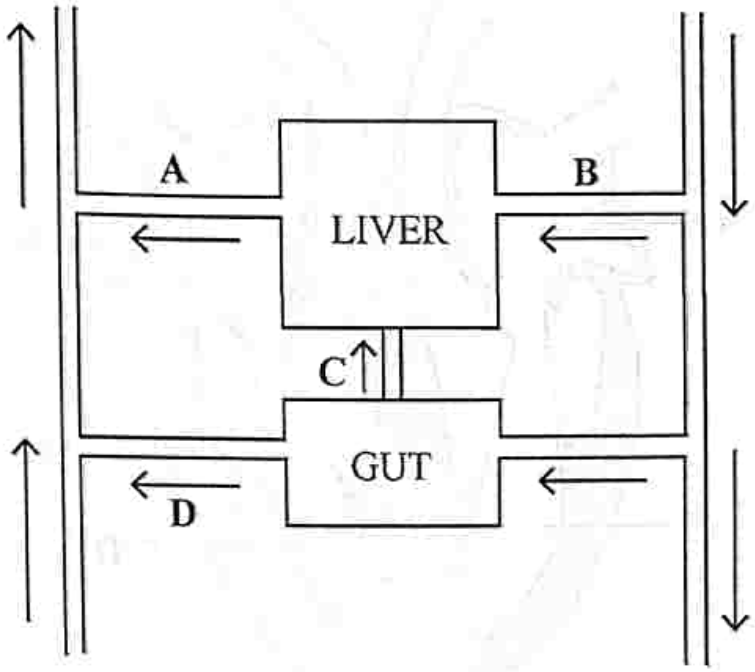
21 The diagram shows the structure of the human skin.

Which part, **A**, **B**, **C** or **D**, acts as an insulator?



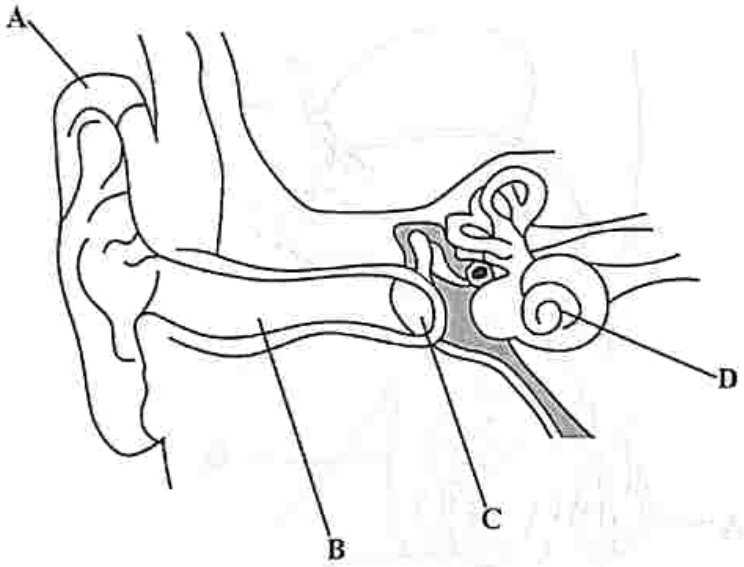
22 The diagram shows the relative positions of the liver, gut and associated blood vessels.

Which blood vessel, **A**, **B**, **C** or **D**, has the highest concentration of glucose?



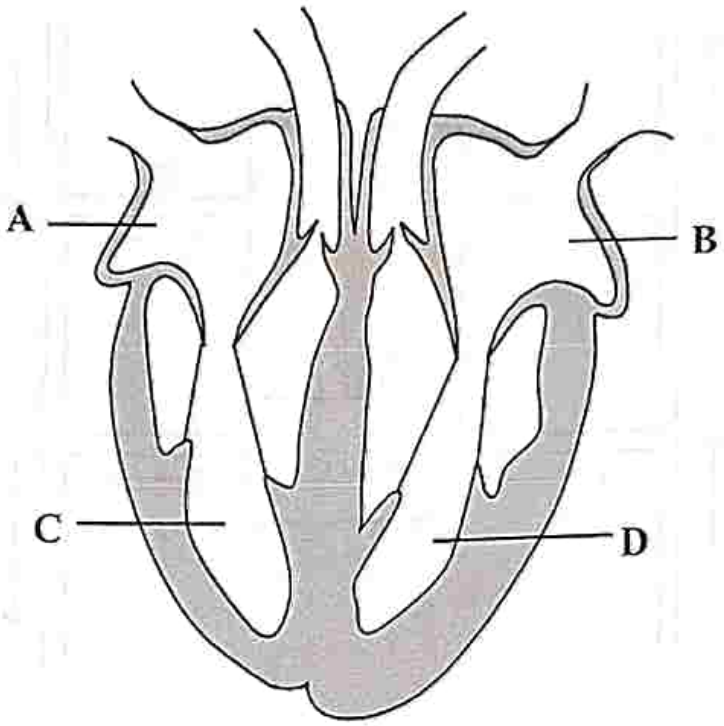
23 The diagram shows the structure of the ear.

Which part, **A**, **B**, **C** or **D**, vibrates due to sound?



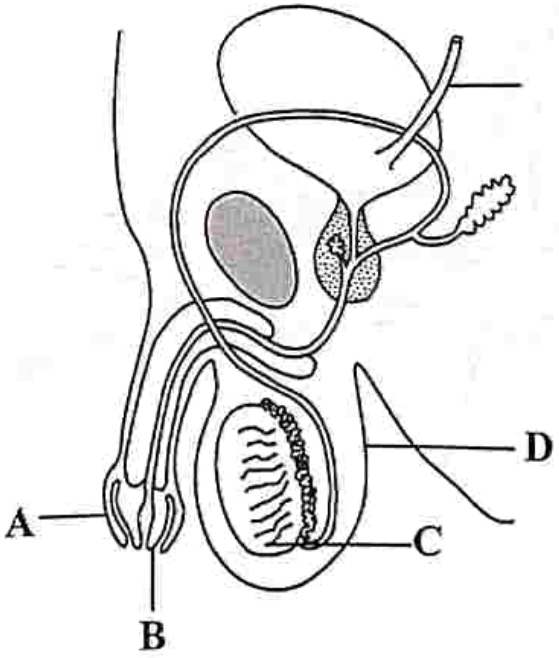
24 The diagram shows a section through the heart.

In which chamber, **A**, **B**, **C** or **D**, is the highest pressure developed during a heart beat?



25 The diagram shows the male reproductive system.

Which part, **A**, **B**, **C** or **D**, is removed during circumcision?



26 Active immunity is a result of

A breastfeeding.
B infection by pathogens.
C injection with antibodies.
D transfer of antibodies.

27 Items (i) to (v) are stages in recombinant DNA technology in random order.

(i) cut out the allele using enzymes
(ii) check that the new allele is acquired
(iii) identify allele for the desired characteristic
(iv) allow the bacterium to reproduce
(v) insert the allele into a bacterium

Which one is the correct sequence of the stages?

A (i), (ii), (iii), (iv), (v)
B (iii), (i), (v), (ii), (iv)
C (ii), (v), (iv), (iii), (i)
D (v), (iv), (iii), (ii), (i)

28 Which statement refers to viruses **only**?

A They contain nucleic acids.
B They reproduce only inside living cells.
C They can only be seen using a microscope.
D They are parasitic and pathogenic.

29 Which cancer is caused mainly by exposure to radiation?

A lung cancer
B skin cancer
C prostate cancer
D breast cancer

30 Albinism is a result of

A dietary deficiencies.
B genetic disorder.
C environmental factors.
D chromosomal abnormality.

31 In cattle, the allele for horns is dominant to that for hornless.
A horned bull was mated with a hornless cow.

Which ratio of offspring is expected if the bull was heterozygous?

A 1 horned : 1 hornless
B 3 horned : 1 hornless
C all hornless
D all horned

32 Which characteristic is an example of continuous variation?

A blood groups
B tongue rolling
C palm length
D ear lobes

33 The following features were identified in an organism:

heterotrophic
non-motile
saprotrophic

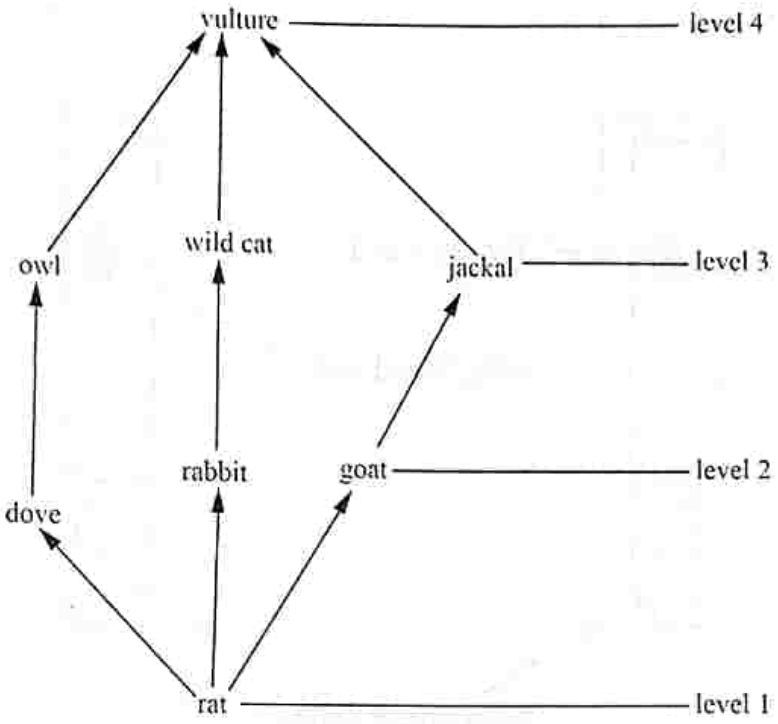
To which kingdom does the organism belong?

A prokaryotae
B protoctista
C fungi
D plantac

34 Which one is **not** a threat to the elephant population in Zimbabwe?

A deforestation
B poaching
C pollution
D culling

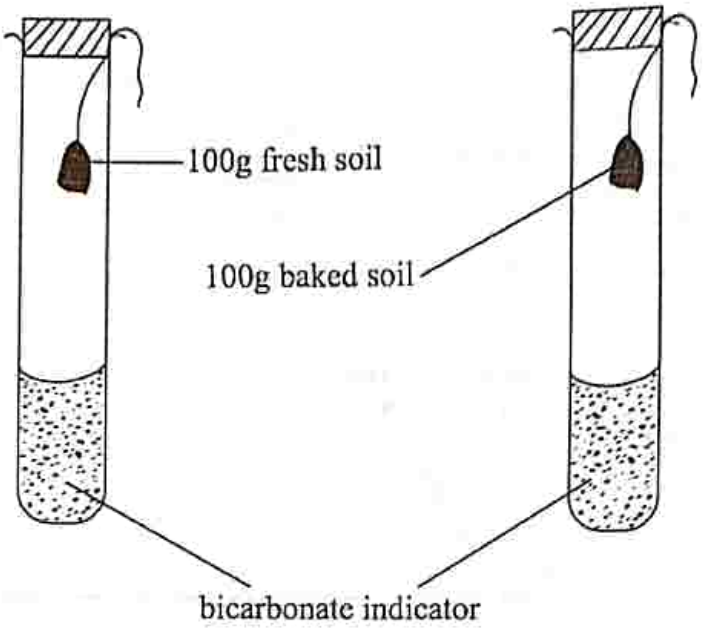
35 The diagram shows a food web.



Which level represents secondary consumers?

A 4
B 3
C 2
D 1

36 An experiment was set up to investigate the presence of microorganisms in soil as shown.



Which change would indicate the presence of microorganisms in soil?

A Bicarbonate indicator changes colour from red to yellow with fresh soil.
B Bicarbonate indicator changes colour from red to yellow with baked soil.
C Bicarbonate indicator changes colour from yellow to red with fresh soil.
D Bicarbonate indicator changes colour from yellow to red with baked soil.

37 Conservation involves

A planting of vegetation on mine dumps.
B mining black granite for export.
C putting anthill soil in sandy fields.
D controlled hunting of wild animals.

38 Which one is a short term effect of alcohol consumption?

A slurred speech
B liver cirrhosis
C loss of employment
D breakdown of families

39 Which infectious disease is caused by a protozoan?

A malaria
B bilharzia
C typhoid
D cholera

40 Which one is **not** a cause of disease?

A microorganisms
B malnutrition
C radiation
D rehydration

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